

# DoangJoo "Alan" Synn

PhD Candidate in Computer Science · Georgia Institute of Technology

Advised by [Prof. HyunJoo Oh](#) and [Prof. Sehoon Ha](#)

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## Research Interests

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My research explores computational design and creativity, with a focus on representing and supporting creative intent in generative and interactive systems. I build models and interfaces that translate high-level human intent into expressive, controllable outcomes, spanning domains such as computer graphics, motion synthesis, and kinematic design.

## Education

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### **Georgia Tech, Ph.D. in Computer Science**

2022 – Present

- Atlanta, GA, USA
- [School of Interactive Computing](#); advised by Profs. [Oh](#) & [Ha](#)
- Research: computational design, automata, and motion synthesis (HCI/Graphics)

### **Korea University, M.Eng. in ECE**

2020 – 2022

- Seoul, Korea
- Advised by [Prof. Jong-Kook Kim](#)
- GPA: 4.25 (4.5 scale)
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

### **Korea University, B.Eng. in EE**

2012 – 2020

- Seoul, Korea
- GPA: 3.56 (4.5 scale)
- Military service leave as a Software Engineer, 2016–2019

## Experience

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### **PhD Research Intern, Dolby Laboratories**

2024 – 2025

- Atlanta, GA, USA
- Experience Delivery Lab, built a time-series, multimodal datastore supporting low-latency media analytics.

### **MLOps Engineering Intern, LG Uplus Corp.**

2023

- Seoul, Korea
- Led **DeepCrunch**, a model-compression library with multi-backend AI inference for efficient edge serving.

### **Research Scientist, Protopia AI Inc.**

2021 – 2022

- Austin, TX, USA
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- CEO: [Dr. Ebrahimi](#); CTO: [Prof. Esmaeilzadeh](#).

## Cloud Architect · AWS Educate Cloud Ambassador

2020 – 2022

- Seoul, Korea
- Led engineering for 4 startup services, translating ambiguous product needs into back-end systems while mentoring early-career engineers.
- Built production cloud infrastructure with Kubernetes, Terraform, CI/CD, and cost-optimized autoscaling.

## Server Engineer, Flysher Inc.

2019

- Seoul, Korea
- Shipped back-end features for 4 social-game titles; one reached global top-10 on Facebook.
- Led an ETL → Redshift analytics pipeline driving data-informed feature rollout.

## System Engineer, AKA Intelligence (Musio)

2017 – 2019

- Seoul, Korea
- Designed embedded Linux boards & firmware for the social robot **Musio**.
- Coordinated HW-SW integration and production QA with overseas ODMs.

## Selected Publications

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- 1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *CHI*, Apr. 2026.
- 2 J. Yu, P. Garg, **D. Synn**, and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *CHI*, Apr. 2025.